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Implementing DPO

Instructor

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DPO: Your Language Model is Secretly a Reward

1. DPO fine-tunes language models by optimizing them directly according to human preferences
2. No need of separate reward model
3. Converts the overall objective into a maximum likelihood problem
4. Memory efficient and training process is relatively more stable

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DPO: Preference Dataset

- Assume you have a preference dataset $\mathcal{D} = \{(x, y_1, y_2, p)\}$ where:

x is the input context.
y_1 and y_2 are two possible outputs for the input x
$p \in \{1,2\}$ indicates the preferred output, i.e., if $p = 1$, humans prefer y_1 over y_2, and vice versa.

- DPO: Model and Scores

The model $\pi_\theta(y|x)$ is parameterized by and gives the probability of output y given input x
For each output y_i, the model assigns a score $s_\theta(x, y_i)$, typically derived from the log-probability:

$$s_\theta(x, y_i) = \log \pi_\theta(y_i|x)$$

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DPO: Objective Function

DPO aims to adjust the model parameters such that the preferred output (according to human feedback) is assigned a higher score

The core idea is to maximize the probability that the preferred output has a higher score than the non-preferred output. This is done by optimizing a binary cross-entropy loss based on the scores.

The probability that the model assigns to the preferred output being better than the non-preferred output can be modeled using a softmax-like function:

$$P_\theta(y_1 > y_2|x) = \frac{\exp(s_\theta(x, y_1))}{\exp(s_\theta(x, y_1)) + \exp(s_\theta(x, y_2))}$$

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DPO: Objective Function

The loss function L_{DPO} for a single pair (x, y_1, y_2, p) is :

$$L_{\text{DPO}}(\theta) = -\log P_\theta(y_p > y_{3-p}|x)$$

where, y_p is the preferred output and y_{3-p} is the non-preferred output

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DPO: Training

The overall objective is to mizimize the average loss over the entire preference dataset:

$$\theta^* = \arg\min_{\theta} \sum_{(x,y_1,y_2,p)\in\mathcal{D}} L_{\text{DPO}}(\theta)$$

DPO: Regularization

To ensure that the model doesn't deviate too far from the pre-trained model, regularization techniques like KL-divergence are often used:

$$L_{\text{DPO}}^{\text{reg}}(\theta) = L_{\text{DPO}}(\theta) + \lambda \cdot \text{KL}(\pi_\theta(\cdot|x) \parallel \pi_{\text{pretrained}}(\cdot|x))$$

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Preference Collection Techniques

- ## Pairwise Comparison

  Present annotators with two outputs and ask them to choose which one they prefer. This is the most common and straightforward method.

- ## Ranking Multiple Outputs

  Provide annotators with more than two outputs and ask them to rank them in order of preference.

- ## Crowdsourcing

  Use platforms like Amazon Mechanical Turk to gather preferences from a diverse group of annotators, increasing the variety of the data.

- ## Contextualized Preferences

  Collect preferences in specific contexts or scenarios to capture more nuanced human judgments, especially when dealing with ambiguous outputs.

- ## Annotation Guidelines

  Provide clear and consistent guidelines to annotators to reduce noise in the preference data.

Thank You